PHARMACEUTICS

UNIT 5 NOTES

SEMI SOLID DOSAGE FORMS

- DEFINITION
- CLASSIFICATION
- FACTORS INFLUENCING DERMAL PENETRATION
- EXCIPIENTS
- EVALUATION

IMPERFECT PHARMACY

CONNECT WITH US ON :

IMPERFECT PHARMACY APP

@IMPERFECTPHARMACY

SEMI - SOLIDS

- Semi solid dosage forms are topical (dermatological) preparations used for therapeutic, protectives or cosmetic function.
- They are generally applied over the skin but can also be applied nasally, vaginally or rectally.
- Pharmaceutical semi solid dosage forms generally include : ointments, pastes, creams and gels.
- They contain one or more active pharmaceutical ingredients (API) dissolved or uniformaly dispersed in a suitable base.

Advantages

- It is used externally hence probability of side effects are very less.
- First pass metabolism is avoided.
- Suitable for unconscious patients.
- Suitable dosage form for bitter drugs.
- More stable than liquid dosage form.

Disadvantages

- No dose accuracy.
- They are bulky to handle
- Application with finger may cause contamination.
- Physio-chemically less stable compared to solid dosage form.
- May cause irritation or allergy in some patients.

IMPERFECT PHARMACY

VIDEOS : IMPERFECT PHARMACY

Ideal Properties of Semi Solid Dosage Forms

It can be defined on the basis of 3 parameters :

- Physical Properties
- Physiological Properties
- Application Properties

Physical Properties

- They should have smooth texture.
- They should be elegant in appearance.
- They should be non-dehydrating.
- They should be non gritty in nature.
- They should have non-greasy and non-staining properties.
- They should be non-hygroscopic in nature.

Physiological Properties

- They should be Non-irritating in nature.
- They should not alter skin membrane
- They should be easily miscible with skin secretion.

Application Properties

- They should be easily applicable with efficient drug release.
- The should have high aqueous washability.

IMPERFECT PHARMACY

VIDEOS : IMPERFECT PHARMACY

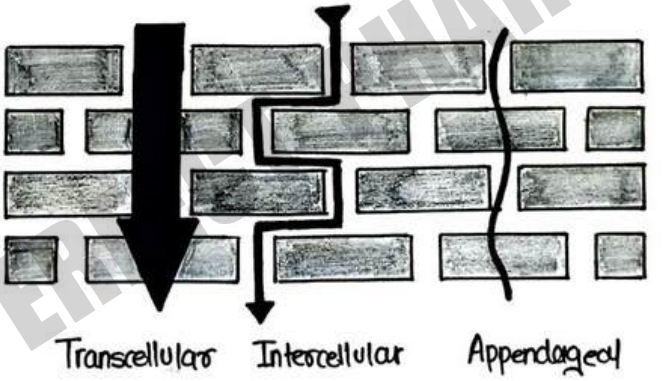
MECHANISM OF DRUG PERMEATION

- It can also be known as Dermal Penetration of Drugs
- The skin has two main layers

① Epidermis (Outermost layer)
② Dermis (Active part of skin containing hair follicles & blood supply)

- Epidermis itself divides into 5 other parts but the main layer that controls the penetration of drugs is called Stratum Corneum.
- The permation / penetration of drugs occurs through three major routes:

(1) Transcellular
(2) Intercellular
(3) Appendageal



IMPERFECT PHARMACY

VIDEOS : IMPERFECT PHARMACY

FACTORS INFLUENCING DERMAL PENETRATION OF DRUGS

These factors can be further divided into two categories

① Biological Factors
② Physiochemical Factors

Biological Factors

Biological factors mainly includes :

- Skin Condition
- Skin Age
- Blood Flow
- Regional Skin Sites
- Skin Metabolism
- Species Difference
- Skin Hydration

Skin Hydration

- When the amount of water is sufficient or in excess the tissue swells and softens.
- Drugs through this softens tissues easily penetrates the skin.
- Hence, Hydration increases drug penetration.

Skin Condition

- The penetration of drugs through skin is affected by Age, Disease, Climate and Injury.
- Drug absorbtion occurs rapidly in children.
- Diseases and Injured skin increases drug penetration
- Drug Penetration is less in healthy skin.

IMPERFECT PHARMACY

VIDEOS : IMPERFECT PHARMACY

Skin Age

- The Young skin is more permeable than older.
- The skin of the childrens are more sensitive to the toxic effects of drugs.
- Skin Age plays an important role in dermal penetration of drugs.

Blood Flow

- Blood flow through the skin also affects the dermal penetration of drugs.
- When blood flow is reduced, amount of drug penetration increases.

Regional Skin Site

- Dermal Penetration of drugs changes with varying of skin thickness.
- Facial skin is generally more permiable compare to other body sites.

Skin Metabolism

- The skin metabolizes steroid harmones, chemical carcinogens and some other drugs.
- So skin metabolism determines efficacy of drug permeated through skin

Species Difference

- The condition and thickness of skin varies in different species.
- Mice, rat, rabbits have more hair follicles but they lack sweat glands, so drug penetration is different in their skin compare to human skin.

IMPERFECT PHARMACY

VIDEOS : IMPERFECT PHARMACY

Physiochemical Factors

Physiochemical factors mainly includes :

- Temperature and pH
- Drug Concentration
- Solubility of Drugs
- Diffusion Coefficient
- Molecular Size and Shape

Temperature and pH

- The increase in temperature increases the dermal penetration of drugs through skin.
- Drug with pH value near 5 easily penetrates the skin.

Drug Concentration

- The drug penetration is depends upon the concentration gradient.
- Concentration gradient will be higher if concentration of drug will be more across the barrier.

Solubility of Drugs

- Highly lipid soluble drugs easily penetrates the skin compare to hydrophilic drugs

Diffusion Coefficient

- The diffusion coefficient of drug depends on the properties of drugs & diffusion medium.

Size and Shape

- Penetration of drugs is inversely depends on size.
- Small molecules penetrates faster than larger ones.

IMPERFECT PHARMACY

VIDEOS : IMPERFECT PHARMACY

THANK YOU

FOR CHOOSING IMPERFECT PHARMACY AS YOUR STUDY PARTNER

IMPERFECT PHARMACY

JOIN US ON :

- IMPERFECT PHARMACY
- IMPERFECT PHARMACY
- @IMPERFECTPHARMA
- IMPERFECT PHARMACY
- @IMPERFECTPHARMACY
- IMPERFECT PHARMACY

IMPERFECTPHARMACY@GMAIL.COM